
SKILLS

- **Languages:** Python, Go, JavaScript, TypeScript, Java, SQL
- **Frameworks & Libraries:** Next.js, React.js, Node.js, Express.js, Angular.js, Flask, Cypress, TailwindCSS
- **Cloud Platforms & Devops:** AWS (Lambda, Route53, SQS, CloudWatch, IAM, EC2, S3, EventBridge), GCP (Cloud Run, BigQuery, Compute Engine), IBM Cloud, Kubernetes, OpenShift, Docker, Git, ArgoCD, Helm
- **Observability:** OpenTelemetry, Honeycomb, Prometheus, Thanos, Grafana, PromQL
- **Distributed Systems:** Kafka, RabbitMQ, gRPC, REST APIs, Microservices, Event-Driven Architecture, DNS
- **Databases:** Redis, Elasticsearch, Neo4J, PostgreSQL, MongoDB
- **AI/ML:** OpenAI SDK, Claude Code SDK, WatsonX, LLMs, Retrieval-Augmented Generation (RAG)
- **Certifications:** [IBM Enterprise Design Thinking Practitioner](#) | [IBM Blockchain Essentials](#) | [IBM Developer Jumpstart](#)

WORK EXPERIENCE

IBM Raleigh, NC
Staff Software Engineer - [NS1 Connect](#) Nov 2025 - Current

- Architected and instrumented metrics, tracing, and OpenTelemetry pipelines across core NS1 Connect services, tracking events from messaging queues through daemon processes and AWS Lambdas to monitor and troubleshoot Route53 synchronization workflows.
- Designed and deployed enterprise Honeycomb dashboards, proactive alerting workflows, and SLO monitors utilized by engineering teams to reduce mean time to detection and streamline production troubleshooting.
- Engineered high-throughput backend enhancements and gRPC services for distributed DNS synchronization workflows, developing features in Go and Python to process lambda-driven events across messaging queues between AWS Route53 and NS1.

Associate Software Engineer - [Hybrid Cloud Mesh Team](#) Aug 2023 - Oct 2025

- Built and enhanced observability microservices integrating Prometheus and Thanos for large-scale metric ingestion, enrichment, and long-term storage, enabling real-time performance insights across multi-cloud and on-premise environments.
- Owned and configured Prometheus remote-write pipelines spanning scraping, middleware enrichment, and Thanos ingestion to enable a unified, topology-aware view of mesh resources via Skupper.
- Designed and built PromMocker, a telemetry simulation framework that automated resource creation and generated realistic Prometheus remote-write traffic, enabling observability validation without deploying full gateway environments.
- Designed and deployed a WatsonX-powered RAG chatbot assistant utilizing custom vector search over internal Mesh documentation to accelerate troubleshooting and reduce time-to-resolution.

Software Engineer Intern - [Cloud Expert Labs](#) May 2022 - Sep 2022

- Built secure authentication UIs using TypeScript, Next.js, and Node.js, integrated with IBM Cloud Identity and IBM Carbon Design.
- Implemented REST APIs with Express to handle secure user authentication flows, including Multi-Factor Authentication (MFA), password resets, and email-based account recovery.
- Automated retailer onboarding workflows with Python, reducing manual processing by ~75% and accelerating customer onboarding.
- Improved product reliability by developing and deploying Cypress E2E test suites from scratch into the CI/CD pipeline, reducing deployment regressions.

Leidos June 2021- Dec 2021
Software Engineer Intern - [Annex Intelligence Division](#) Reston, VA

- Developed secure web and mobile applications for the National Counterterrorism Center (NCTC), building user-facing features that improved data accessibility and operational workflows for critical intelligence initiatives.
- Optimized backend database lookup and retrieval speeds by refactoring complex Elasticsearch indexing and query execution patterns.
- Streamlined CI/CD pipelines using Docker, GitLab, and SonarQube, accelerating automated build and deployment cycles by 40%.

RECENT PROJECTS

Peerfolio | [Github](#) Jun 2025
[Peerfolio.org](#) is a social investment platform in development that integrates with Plaid to aggregate brokerage accounts, enabling users to track portfolios, compare with friends, and explore investing trends.

YTRecap | [Github](#) Mar 2023
[YTRecap.org](#) is a web app built using Python Flask that can summarize any YouTube video by parsing the video for its closed captions and passing them to an NLP large language model

EDUCATION

University of Pittsburgh May 2023
Bachelor of Computer Science (Honors) Pittsburgh, PA
Coursework: Algorithmic Implementation, Data Structures, Discrete Structures, Assembly Language, Intermediate Programming, Computer Organization, Software Engineering, Web-Development, Artificial Intelligence, Systems Software, Formal Methods in Computing, Operating System, Database Management Systems, Quality Assurance, Big Ideas in Computing